

Amir Esmaeili

AI/ML ENGINEER | BACKEND SYSTEMS | PRODUCTION ML

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Location: Tabriz, Iran (Open to relocation)

Professional Summary

AI/ML Engineer with backend expertise, specializing in deploying production-grade machine learning systems and scalable APIs. Delivering measurable business impact through scalable ML solutions and high-performance backend systems.

Work Experience

ML Engineer/ Data Scientist *Desmer Guvenlik* [🔗](#)

02/2024 – 03/2026

- Engineered and deployed AI optimization models for logistics operations, reducing service delays by 30% and cutting operational costs across a network of 100+ service points.
- Applied causal inference techniques to evaluate business process changes, directly contributing to a 20% reduction in service delivery delays.
- Integrated ML models into live production systems, enabling real-time decision-making and reducing manual oversight by 40% across daily operations.
- Conducted demand forecasting on ATM cash withdrawals across 50+ locations during seasonal periods, optimizing resource allocation and reducing cash shortages by 25%

Backend Developer *Aral Studio*

02/2023 – 01/2024

- Designed and optimized backend systems capable of handling high user traffic in real time, ensuring excellent uptime and system reliability under peak loads.
- Improved API response time by 35% through codebase refactoring and strategic database indexing across 10+ high-traffic endpoints.
- Architected microservices infrastructure serving multiple product lines, reducing deployment time by 40% and improving team scalability.

Backend Developer *NOAY*

09/2021 – 01/2023

- Built secure and scalable backend infrastructure for a cryptocurrency platform, supporting high-frequency transactions with near-zero failure rates.
- Developed robust APIs and microservices that reduced server load by 25% and improved overall system responsiveness.
- Partnered with frontend teams to ensure seamless integration and intuitive user experience, increasing user satisfaction scores.

Qualifications

Programming	Python, PHP, JS, C++
ML/AI	PyTorch, TensorFlow, HuggingFace, LangChain, OpenAI API
Backend	Django, Laravel, REST APIs, Microservices
Data & DB	PostgreSQL, MySQL, Redis, pgvector
Tools	Docker, Linux, Git, Celery
Languages	English – IELTS: 7, Persian, Turkish, Azarbaijani
Soft Skills	Problem-solving, Critical thinking, Research methodology, Scientific writing, Teamwork and collaboration, Project management

Projects

Resume Matching System *Personal Project*

07/2025 – Present

- Built an AI-powered matching system using text embeddings to align candidate profiles with job descriptions.
- Designing and implementing the backend using Django and PostgreSQL with pgvector for vector similarity search.
- Integrating Hugging Face models for text embedding and OpenAI for job description enhancement via chatbot.
- Architecting modular APIs and background job processing using Celery and Redis.
- Emphasizing scalable deployment with Docker and secure API access.
- Skills:** Python, Django, pgvector, Hugging Face Transformers, OpenAI API, Celery, Redis, Docker, Local LLM

Question Generating App *Freelance Project*

07/2025 – 08/2025

- Developed a prompt-engineered question generation platform using OpenAI, Gemini, and DeepSeek APIs tailored for educational institutes.
- Enabled teachers to upload resources (PDFs, text, images) for automatic exam question generation across cognitive levels and subject domains.
- Supported rich outputs including diagrams, LaTeX equations, and images using multimodal API capabilities.
- Implemented RESTful APIs and handled file preprocessing, metadata extraction, and custom prompt templates for accuracy.
- Emphasized modular pipeline structure with logging, error handling, and scalability for concurrent users.
- **Skills:** Python, OpenAI API, Gemini API, DeepSeek API, Prompt Engineering, REST APIs, File Handling

Insurance Recommender System *Freelance Project*

03/2025 – 05/2025

- Developed a recommender system analyzing customer purchase history and behavioral patterns to predict insurance preferences; directly supporting customer segmentation and personalized product recommendations similar to marketing campaign targeting.
- Implemented a hybrid recommender system combining collaborative filtering and content-based filtering approaches.
- Analyzed customer demographics, purchase history, and behavioral interactions to identify cross-sell opportunities and predict customer lifetime value (CLV) - directly aligned with marketing analytics practices.
- Deployed a real-time insurance recommendation engine via REST APIs, driving personalized product offers and improving conversion potential.
- Conducted testing simulations and evaluated metrics to tune the recommendation quality.
- **Skills:** Python, Scikit-learn, Pandas, Recommender Systems, Feature Engineering, REST APIs, Testing

AI-Powered Itinerary Generator *Personal Project*

08/2025

- Developed a serverless application that generates detailed travel itineraries using the OpenAI GPT-4o model.
- Architected an asynchronous backend using Cloudflare Workers that immediately returns a job ID and processes requests in the background.
- Implemented a SvelteKit frontend to provide real-time status updates by listening to a Google Firestore database.
- Ensured data integrity through schema validation with Zod and robust data parsing.
- **Skills:** Cloudflare Workers, SvelteKit, Google Firestore, OpenAI API, Zod, Node.js

Educational Chatbot *Personal Project*

05/2025

- Built a backend service to answer educational questions based on provided content using natural language processing.
- Implemented a three-step process of content embedding, retrieval, and answer generation.
- Utilized a multilingual sentence transformer model for creating vector embeddings of the educational content.
- Deployed the application using Docker for portability and scalability.
- **Skills:** Python, FastAPI, Transformers, Docker

Education

M.Sc. Soft Computing and Artificial Intelligence *University of Tabriz*

09/2023 – present

- Selected as a "Brilliant Talent", granted direct admission to M.Sc. without National University Entrance Exam
- Awarded a full scholarship for M.Sc. studies, covering all educational expenses
- Selected Specialized Course Grades: Artificial Intelligence: A, Data Mining: A+, Machine Learning: B, Deep Learning: B+

B.Sc. Computer Science *University of Tabriz*

09/2019 – 06/2023

- Full Scholarship Recipient
- Ranked 4th in Overall B.Sc. GPA
- Honorable Mention in The International Collegiate Programming Contest - 2019, 2020, 2021
- Top 3% in National University Entrance Exam
- Selected Specialized Course Grades: Advanced Programming: A+, Data Structures: A+, Probability: A+, Discrete mathematics: A+, Linear algebra: A, Algorithms: A+, Theory of Computation: A+, Compiler: A+

High School Diploma *National Organization for Development of Exceptional Talents (Sampad)*

2014 – 2018